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## B. Tech. Degree VIII Semester Examination April 2016

### ME 1801 COMPRESSIBLE FLUID FLOW (2012 Scheme)

Time: 3 Hours

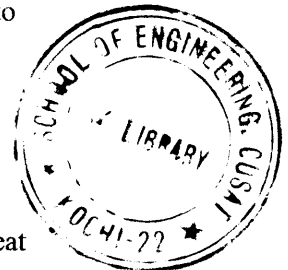
Maximum Marks: 100

(Use of approved gas tables allowed)

#### PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Briefly explain various regions of flow with the help of a steady flow ellipse.
- (b) Define impulse function. Derive an expression for impulse function in terms of Mach number.
- (c) Derive an expression for entropy change across a normal shock wave.
- (d) Briefly explain two dimensional plane shock wave. Where are these types of waves found?
- (e) Compare fanno curves for different values of mass flow density with the help of h-s diagram.
- (f) In Rayleigh flow, when is the maximum possible heat transfer reached? Write an expression for maximum heat transfer ( $Q_{\max}$ ). What will happen to  $Q_{\max}$  when
  - (i)  $M = 0$
  - (ii)  $M = 1$
  - (iii)  $M = \alpha$  ?
- (g) Briefly explain Rayleigh's supersonic pitot formula.
- (h) Explain the principle and working of shock tube with the help of a neat sketch.



#### PART B

(4 × 15 = 60)

- II. (a) Briefly explain various regions of atmosphere. (9)
  - (b) An air stream at pressure 1 bar, temperature 400 K and velocity 400 m/s is brought to rest (6)
    - (i) adiabatically
    - (ii) isentropically.
 Determine the stagnation pressure and temperature in both cases.
- OR**
- III. A nozzle in a wind tunnel gives a test section Mach number of 2. Air enters the nozzle from a large reservoir at 0.69 bar and 310 K. The cross sectional area of the throat is 1000 cm<sup>2</sup>. Determine the following quantities for one dimensional isentropic flow. (15)
    - (i) Pressure, temperature and velocities at the throat and test sections.
    - (ii) Area of cross section of the test section.
    - (iii) Mass flow rate.

(P.T.O.)

- IV. The stagnation pressure and temperature of air at entry of a nozzle are 5 bar and 500 K respectively. The exit Mach number is 2.0 where a normal shock occurs. Calculate the following quantities before and after the shock; (i) static and stagnation pressures and temperatures (ii) air velocities and (iii) Mach numbers. What are the values of stagnation pressure loss and increase in entropy across the shock? (15)

**OR**

- V. (a) Derive Prandtl's equation for an oblique shock wave. (8)  
 (b) A gas ( $\gamma = 1.3$ ) at pressure 345 mbar, temperature 350 K and Mach number 1.5 is to be isentropically expanded to 138 mbar. Determine the deflection angle. (7)

- VI. Air at stagnation pressure 10 bar and stagnation temperature 400 K, is supplied to a 50 mm diameter pipe. The friction factor for the pipe surface is 0.002. If the Mach number changes from 3.0 at the entry to 1.0 at the exit, determine (15)  
 (i) the length of the pipe  
 (ii) mass flow rate

**OR**

- VII. A combustion chamber in a gas turbine plant receives air at 350 K, 0.55 bar and 75 m/s. The air fuel ratio is 29 and the calorific value of the fuel is 41.87 MJ/kg. Taking  $\gamma = 1.4$  and  $R = 0.287$  kJ/kg K for the gas, determine: (15)  
 (i) the initial and final Mach number.  
 (ii) final pressure, temperature and velocity of the gas.  
 (iii) percentage stagnation pressure loss in the combustion chamber  
 (iv) the maximum stagnation temperature attainable.

- VIII. Explain the working of the following with help of neat sketches. (15)  
 (i) Schlieren apparatus  
 (ii) Dual beam laser doppler velocimeter  
 (iii) Hot wire anemometer

**OR**

- IX. Explain the working principles of the following with the help of neat sketches. (15)  
 (i) Pitot tube  
 (ii) Supersonic wind tunnel  
 (iii) Interferometer